REVISION 2021 • SEMESTER IV • ELECTRICAL AND ELECTRONICS ENGINEERING

Electrical Installation Design and Estimation

A complete diploma handbook for Course 4032. It turns the supplied official syllabus into a practical learning and exam-preparation resource for all modules, learning supports, and exam preparation materials as per the official syllabus.

COURSE CODE
4032

SEMESTER
IV

TYPE
Theory

CREDITS
4

TEACHING SCHEME
4:0:0:0

CONTACT HOURS
60

ASSESSMENT
CIE 40 + ESE 60

MODULES
4

Official source declaration and course dashboard

Course facts, module coverage, activities and assessment data below are transcribed from the supplied official course PDF. Teaching explanations, diagrams, practice questions and model paper are POLY PMNA study support, not an official examination paper.

Source-integrity note

The user confirmed this source for placement in the Revision 2021 lesson library. However, each supplied PDF page footer reads “Diploma Curriculum Revision 2026”. This handbook therefore uses Revision 2021 for the website label and file location as instructed, while preserving the discrepancy transparently. Students must confirm their active curriculum scheme with their institution or SITTTR before relying on a syllabus for examination decisions.

Confirmed course facts

Official course information

Item	Value
Programme(s)	Electrical and Electronics Engineering; Civil & Rural Engineering; Civil & Environmental Engineering; Civil Engineering (Construction Technology)
Course title	Electrical Installation Design and Estimation
Course code	4032
Semester	IV
Course type	Theory
Teaching scheme L:T:P:R	4:0:0:0
Contact hours	60
Credits	4
CIE / ESE	40 / 60
Self-learning	6 hours per week; 90 hours total
Prerequisites	Chemistry for Engineering Practices (2003A); Fundamentals of Building Construction (1011)

Course objectives

1. Understand classification, properties, manufacture, testing and suitability of stone, timber, lime, bricks, blocks and alternative materials.
2. Identify finishing materials, wood products, glass, metals, special materials, industrial/agro waste

products and structural steel sections.

3. Become familiar with prefabrication, site layout, earthwork and foundation practices.

4. Understand the types, components, fittings and functional requirements of openings, roofs, trusses and vertical communication systems.

Course outcomes

Official course outcomes

CO	Outcome	Bloom level
CO1	Explain classification, properties, uses, manufacture, testing and suitability of basic building materials.	Understand
CO2	Identify appropriate finishing materials, wood products, glass, metals and special materials.	Understand
CO3	Describe prefabrication, site layout, earthwork operations and foundation types.	Understand
CO4	Explain types, components and functional requirements of doors, windows, ventilators, roofs, trusses and vertical communication systems.	Understand

CO–PO mapping

Official articulation matrix

Course outcome	PO1	PO2	PO3–PO11
CO1	3	2	–
CO2	3	2	–
CO3	3	2	–
CO4	3	–	–

Legend: 1 = low correlation, 2 = medium correlation and 3 = high correlation. The PDF totals are PO1 = 12 and PO2 = 6.

Syllabus roadmap

Use the hours as a guide for study effort. A short module is not a reason to skip its drawing, selection logic or common mistakes.

Module plan based on the official major topics

Module	CO	Hours	Focus	Study evidence

Module	CO	Hours	Focus	Study evidence
M1: Basic building materials	CO1	18	Classification of materials; Stone; Stone quality; Quarrying and dressing of stone; Timber; Seasoning of timber; Defects and bamboo; Lime; Brick earth and traditional bricks; Burnt clay bricks; Fly ash bricks and AAC blocks	Explain, select, sketch, compare and apply.
M2: Flooring, finishing and special materials	CO2	15	Flooring materials; Wood products; Glass; Metals; Insulating and protective materials; Plaster of Paris; Industrial and agro-waste materials; Special processed materials	Explain, select, sketch, compare and apply.
M3: Building construction practices and foundation	CO3	12	Prefabricated structures; Structural steel buildings; Substructure and site layout; Setting out load-bearing structures; Setting out framed structures; Earthwork; Plinth filling; Foundations; Pile foundations; Well foundations	Explain, select, sketch, compare and apply.
M4: Building components and roofing	CO4	15	Doors; Doors and shutters; Windows; Window forms; Stairs; Lifts and escalators; Roofs and roof trusses	Explain, select, sketch, compare and apply.

M1 study route

Classify materials and make sound first-stage choices for stone, timber, lime, bricks, blocks and alternative masonry units.

- ✓ Classify natural, artificial, special, finished and recycled materials.
- ✓ Recognise properties that make a material suitable for a construction duty.
- ✓ Connect raw material, manufacture, test and field use.

M2 study route

Select surface, board, glass, metal, insulating and special materials by service condition rather than appearance alone.

- ✓ Compare flooring and finish options for function, wear and maintenance.
- ✓ Identify common wood products, glass and metal uses.
- ✓ Recognise insulating, waste-derived and special processed materials.

M3 study route

Move from material knowledge to the organised construction of a safe substructure, including layout, excavation and foundation choice.

- ✓ Explain prefabrication and basic structural-steel building methods.
- ✓ Follow the logical sequence from site clearance to setting out and excavation.
- ✓ Choose a foundation concept from ground and loading conditions.

M4 study route

Understand building openings, movement systems and roofs as functional assemblies that must control access, light, air, water and load.

- ✓ Identify common doors, windows, ventilators and their functions.
- ✓ Explain stairs, lifts and escalators at an introductory building-components level.
- ✓ Compare flat and pitched roofs and recognise roof-truss terminology.

M1 • CO1 • 18 HOURS

Basic building materials

Classify materials and make sound first-stage choices for stone, timber, lime, bricks, blocks and alternative masonry units.

Learning goals

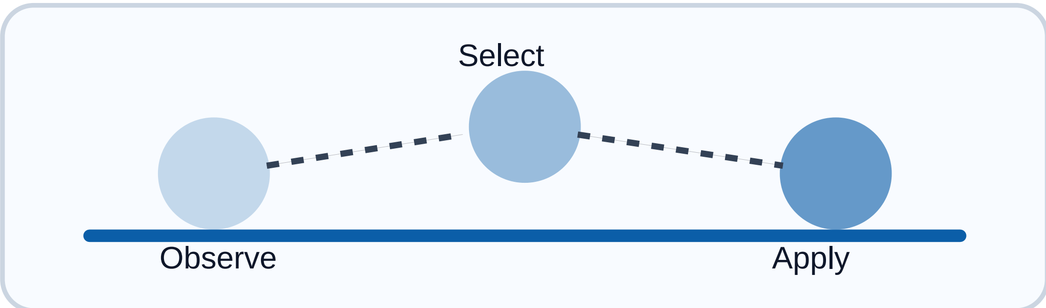
- ✓ Classify natural, artificial, special, finished and recycled materials.
- ✓ Recognise properties that make a material suitable for a construction duty.
- ✓ Connect raw material, manufacture, test and field use.

Engineering habit

Observe the service condition first. Then identify material or component properties, construction detail, inspection point, safety condition and maintenance consequence.

Exam habit

Start with the official term, define it clearly, classify or list the required points, use a labelled sketch where useful, and conclude with a relevant construction application.



Use the sequence: observe the requirement, select the suitable component or material, then apply it with correct site practice.

M1.1 — Classification of materials

Classification of materials

> Official syllabus focus: Types: natural, artificial, special, finished and recycled.

Classify a material first by origin and then by performance. Natural materials such as stone and timber need selection and processing; artificial materials such as bricks and blocks are manufactured to a controlled form. Finished or special products are selected for a particular function such as protection, appearance or strength.

Study method

When comparing options for a wall, record the source, expected exposure, weight, durability and availability before choosing it.

Construction application

Material classification tells a site team what inspection, storage and workmanship controls may be required.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain classification of materials in your own words, give one suitable construction use, and state one inspection or safety point.

M1.2 — Stone

Stone

> Official syllabus focus: Classification of rocks: igneous, sedimentary and metamorphic.

Rock origin affects texture, structure and likely behaviour. Igneous rocks form from cooling molten material, sedimentary rocks form from deposited layers, and metamorphic rocks are altered by heat and pressure. A civil technician should not assume that every hard-looking stone has equal resistance to water, weathering or load.

Study method

Observe bedding, grain, cracks and water absorption before accepting stone for a structural or facing purpose.

Construction application

Correct rock identification supports appropriate use in foundations, masonry, aggregates and architectural finishes.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain stone in your own words, give one suitable construction use, and state one inspection or safety point.

M1.3 — Stone quality

Stone quality

> Official syllabus focus: Requirements of good building stone; general characteristics; laterite stones.

A good building stone should be strong enough for its duty, durable in the local environment, reasonably uniform, free from injurious cracks and able to resist water and weathering. Laterite is common in Kerala and needs careful selection because its condition and protection strongly influence long-term performance.

Study method

Ask whether the stone will remain dry, stay in contact with soil, receive repeated wetting or become an exposed face.

Construction application

Selecting a sound stone reduces cracking, spalling and avoidable maintenance in walls and foundations.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain stone quality in your own words, give one suitable construction use, and state one inspection or safety point.

M1.4 — Quarrying and dressing of stone

Quarrying and dressing of stone

> Official syllabus focus: Quarrying and dressing of stone.

Quarrying removes stone from its natural bed without creating unnecessary cracks. Dressing brings the stone to the shape and surface needed for masonry. The important idea is that handling, cutting and bedding direction can affect both appearance and performance.

Study method

Plan the required sizes before dressing; reject pieces with hidden cracks or weak laminated planes.

Construction application

Good dressing improves joint quality and reduces excessive mortar thickness in stone masonry.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain quarrying and dressing of stone in your own words, give one suitable construction use, and state one inspection or safety point.

M1.5 — Timber

Timber

> Official syllabus focus: Structure of timber; properties and uses of good timber.

Timber is a natural anisotropic material: its behaviour differs along and across the grain. A good timber member is appropriately seasoned, durable for its exposure, sufficiently strong, workable and free from serious defects. It is used where its strength-to-weight ratio, appearance or workability is valuable.

Study method

Identify grain direction, knots, decay marks and moisture condition before using timber for a frame, shutter, door or temporary work.

Construction application

Understanding timber structure helps prevent distortion, splitting and premature decay.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain timber in your own words, give one suitable construction use, and state one inspection or safety point.

M1.6 — Seasoning of timber

Seasoning of timber

> Official syllabus focus: Different methods of seasoning of timber.

Seasoning reduces moisture to a level suitable for service. Controlled drying improves dimensional stability, decreases the risk of fungal attack and supports better finishing. The method must reduce moisture without producing excessive checks, warp or case hardening.

Study method

Stack boards with spacers for air circulation and protect the stack from rain and direct ground moisture.

Construction application

Proper seasoning is essential for joinery, doors, windows and stable timber components.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain seasoning of timber in your own words, give one suitable construction use, and state one inspection or safety point.

M1.7 — Defects and bamboo

Defects and bamboo

> Official syllabus focus: Defects in timber; use of bamboo in construction.

Defects may arise from growth, conversion, seasoning or biological attack. Knots, shakes, warping and decay reduce suitability depending on location and load. Bamboo is a rapidly renewable material that can be useful in low-cost and rural construction when it is selected, treated, detailed and protected from moisture and pests.

Study method

Do not hide a defect under paint; assess whether it affects a critical zone such as a joint, bearing or exposed edge.

Construction application

Defect recognition and responsible bamboo use support safer, lower-waste construction.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain defects and bamboo in your own words, give one suitable construction use, and state one inspection or safety point.

M1.8 — Lime

Lime

Official syllabus focus: Lime: types and uses.

Lime is used in mortar, plaster and soil-related construction applications according to its type and hydraulic behaviour. Its usefulness depends on workability, compatibility with masonry, curing and the required resistance to moisture.

Study method

Choose lime only after checking the service condition and the specified mix; curing is part of performance, not an optional finishing step.

Construction application

Lime remains relevant for mortars, plasters and repair work where compatibility and breathability matter.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain lime in your own words, give one suitable construction use, and state one inspection or safety point.

M1.9 — Brick earth and traditional bricks

Brick earth and traditional bricks

> Official syllabus focus: Constituents of brick earth; conventional or traditional bricks; nominal and standard size.

Brick earth must contain constituents in suitable proportion so that moulded bricks dry and burn without excessive cracking or distortion. Standard and nominal dimensions help masons plan bonding, joints and wall dimensions correctly.

Study method

Check dimensional consistency, edges, sound on striking and visible cracks during a simple field inspection.

Construction application

Uniform brick size improves bond, alignment, material estimation and finish quality.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain brick earth and traditional bricks in your own words, give one suitable construction use, and state one inspection or safety point.

M1.10 — Burnt clay bricks

Burnt clay bricks

> Official syllabus focus: Classification of burnt clay bricks and their suitability; manufacturing process of burnt clay bricks.

Burnt clay bricks are classified according to quality and intended use. Manufacturing generally involves preparation of earth, moulding, drying and burning. Each stage affects strength, porosity, shape and colour; poor drying or burning can produce weak or distorted units.

Study method

Relate the expected wall exposure and load to the brick class instead of choosing by colour alone.

Construction application

Knowledge of manufacture helps diagnose defects and select bricks for masonry, partition or facing work.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain burnt clay bricks in your own words, give one suitable construction use, and state one inspection or safety point.

Fly ash bricks and AAC blocks

> Official syllabus focus: Fly ash bricks and autoclaved aerated concrete blocks: brief description only.

Fly ash bricks use industrial by-product material under controlled manufacture. AAC blocks are lightweight aerated units with thermal and handling advantages. Both require compatible mortar, careful cutting and detailing appropriate to the product rather than assumptions based on conventional brickwork.

Study method

Verify manufacturer guidance, unit condition and the specified jointing material before use.

Construction application

Alternative units can improve productivity and resource efficiency when selected for the correct wall system.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain fly ash bricks and aac blocks in your own words, give one suitable construction use, and state one inspection or safety point.

M1 closing checklist

- ✓ I can classify natural, artificial, special, finished and recycled materials.
- ✓ I can recognise properties that make a material suitable for a construction duty.
- ✓ I can connect raw material, manufacture, test and field use.
- ✓ I can draw or describe one site-relevant application and one likely mistake.

M2 • CO2 • 15 HOURS

Flooring, finishing and special materials

Select surface, board, glass, metal, insulating and special materials by service condition rather than appearance alone.

Learning goals

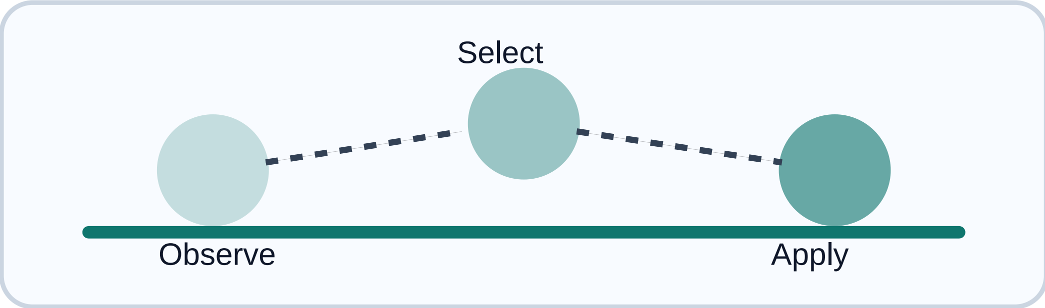
- ✓ Compare flooring and finish options for function, wear and maintenance.
- ✓ Identify common wood products, glass and metal uses.
- ✓ Recognise insulating, waste-derived and special processed materials.

Engineering habit

Observe the service condition first. Then identify material or component properties, construction detail, inspection point, safety condition and maintenance consequence.

Exam habit

Start with the official term, define it clearly, classify or list the required points, use a labelled sketch where useful, and conclude with a relevant construction application.



Use the sequence: observe the requirement, select the suitable component or material, then apply it with correct site practice.

M2.1 — Flooring materials

Flooring materials

> Official syllabus focus: Natural and artificial flooring: granite, marble, clay tiles, ceramic tiles and pavement blocks.

Flooring selection is a performance decision. Wear, slip resistance, water exposure, cleaning method, impact, appearance and repairability all matter. Granite and marble are natural stone finishes; tiles and pavement blocks offer manufactured alternatives with different joint, laying and maintenance requirements.

Study method

Match the floor to the room: a wet area, corridor, workshop and outdoor pavement do not demand the same surface.

Construction application

Correct flooring selection improves safety, durability and maintenance cost.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain flooring materials in your own words, give one suitable construction use, and state one inspection or safety point.

M2.2 — Wood products

Wood products

Official syllabus focus: Plywood, particle board, veneers, laminated board and their uses.

Engineered wood products make efficient use of timber by arranging veneers, particles or laminations for particular panel properties. They are not interchangeable: moisture exposure, edge sealing, fixing method and load direction must be considered.

Study method

Read the product grade and protect cut edges where moisture may enter.

Construction application

Wood products are widely used in partitions, furniture, formwork and interior joinery.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain wood products in your own words, give one suitable construction use, and state one inspection point or safety point.

M2.3 — Glass

Glass

Official syllabus focus: Soda lime, lead, toughened and borosilicate glass: types and uses.

Glass type is selected by optical, thermal, safety or chemical-performance needs. Soda-lime glass is common, lead glass has specialised applications, toughened glass is strengthened for safety-related use, and borosilicate glass has improved resistance to thermal shock.

Study method

Specify safe handling, correct edge treatment and suitable framing; glass failure can be sudden.

Construction application

Knowing glass type helps choose safer windows, partitions, glazing and specialised components.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain glass in your own words, give one suitable construction use, and state one inspection or safety point.

M2.4 — Metals

Metals

> Official syllabus focus: Ferrous and non-ferrous metals and their uses.

Ferrous metals contain iron and commonly provide structural strength, while non-ferrous metals such as aluminium offer corrosion resistance or low mass for particular applications. Selection must consider corrosion, jointing, coating, thermal movement and galvanic interaction.

Study method

Do not treat every shiny metal as corrosion-proof; exposure and contact with other metals matter.

Construction application

Metal choice affects structural members, frames, roofing, fittings and service life.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain metals in your own words, give one suitable construction use, and state one inspection or safety point.

M2.5 — Insulating and protective materials

Insulating and protective materials

> Official syllabus focus: Waterproofing, termite proofing, thermal and sound insulating materials: types and suitability.

Protective materials control unwanted water, pests, heat or sound. The correct material is only one part of a system: continuity, laps, joints, substrate preparation and protection from damage determine whether it works in service.

Study method

Trace the path of water, heat, termites or noise before selecting a protective layer.

Construction application

A well-detailed protective system prevents costly hidden damage in a building.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain insulating and protective materials in your own words, give one suitable construction use, and state one inspection or safety point.

M2.6 — Plaster of Paris

Plaster of Paris

> Official syllabus focus: Constituents and uses of POP and POP finishing boards.

Plaster of Paris is a quick-setting gypsum-based finishing material used for smooth surfaces, mouldings and boards. Its setting behaviour demands correct mixing, timely application and dry-service detailing.

Study method

Prepare only a workable quantity; once setting begins, do not restore workability by uncontrolled addition of water.

Construction application

POP is useful for interior finishing where a smooth controlled surface is required.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain plaster of paris in your own words, give one suitable construction use, and state one inspection or safety point.

M2.7 — Industrial and agro-waste materials

Industrial and agro-waste materials

> Official syllabus focus: Industrial waste: fly ash, blast furnace slag; agro-waste: rice husk, bagasse and coir fibres.

Waste-derived materials can reduce demand for virgin resources when their properties and limits are understood. Fly ash and slag may contribute to construction products, while rice husk, bagasse and coir fibres can be used in suitable composites or low-impact products.

Study method

Describe both benefit and limitation: availability alone does not prove structural suitability.

Construction application

Responsible use can support material efficiency and locally relevant construction solutions.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain industrial and agro-waste materials in your own words, give one suitable construction use, and state one inspection or safety point.

M2.8 — Special processed materials

Special processed materials

> Official syllabus focus: Ferrocrete, artificial timber, MDF, HDF, PVC foam board, WPC, ACP and manufactured sand: brief description and uses.

Special materials are made to obtain a targeted combination of form, finish, durability, workability or resource efficiency. MDF/HDF are interior boards with different densities; WPC combines wood-like filler with polymer; ACP is a composite panel; manufactured sand is processed aggregate for concrete and mortar when properly graded.

Study method

Use a product only within its intended environment and fastening system; a finish panel is not automatically a structural member.

Construction application

Special materials expand design options but require specification awareness.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain special processed materials in your own words, give one suitable construction use, and state one inspection or safety point.

M2 closing checklist

- ✓ I can compare flooring and finish options for function, wear and maintenance.
- ✓ I can identify common wood products, glass and metal uses.
- ✓ I can recognise insulating, waste-derived and special processed materials.
- ✓ I can draw or describe one site-relevant application and one likely mistake.

M3 • CO3 • 12 HOURS

Building construction practices and foundation

Move from material knowledge to the organised construction of a safe substructure, including layout, excavation and foundation choice.

Learning goals

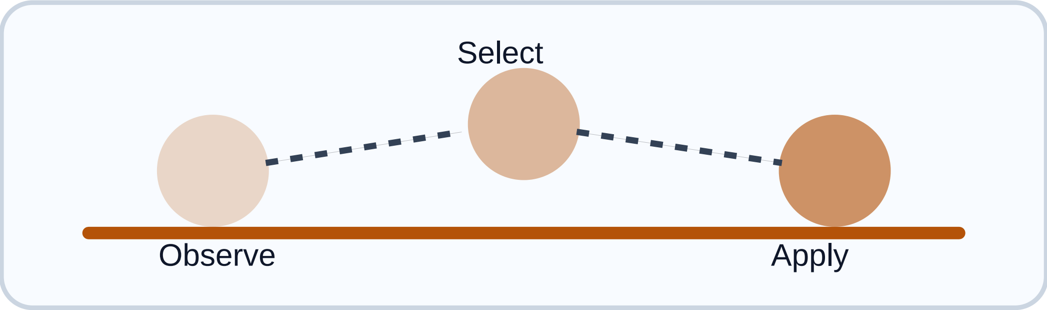
- ✓ Explain prefabrication and basic structural-steel building methods.
- ✓ Follow the logical sequence from site clearance to setting out and excavation.
- ✓ Choose a foundation concept from ground and loading conditions.

Engineering habit

Observe the service condition first. Then identify material or component properties, construction detail, inspection point, safety condition and maintenance consequence.

Exam habit

Start with the official term, define it clearly, classify or list the required points, use a labelled sketch where useful, and conclude with a relevant construction application.



Use the sequence: observe the requirement, select the suitable component or material, then apply it with correct site practice.

M3.1 — Prefabricated structures

Prefabricated structures

> Official syllabus focus: Brief description of prefabricated structures.

Prefabrication means components are manufactured or assembled under controlled conditions before being transported and erected at site. Benefits may include speed, repeatability and reduced wet work, but success depends on tolerances, lifting, connections, transport and sequencing.

Study method

Think of the joint and erection route before approving a prefabricated component.

Construction application

Prefabrication supports faster construction when design and site coordination are strong.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain prefabricated structures in your own words, give one suitable construction use, and state one inspection or safety point.

M3.2 — Structural steel buildings

Structural steel buildings

> Official syllabus focus: Construction methods for structural steel buildings.

Structural-steel construction involves setting out, foundations and base connections, fabrication, transport, erection, alignment, bolting or welding, protection and inspection. Stability during erection is critical because the final load path may not exist until bracing and connections are complete.

Study method

Plan temporary stability, lifting sequence and inspection of connections as part of the method.

Construction application

Steel systems are used for long spans, industrial buildings and fast-track frames.

Technician check

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Common mistake to avoid

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One-minute recall: Explain structural steel buildings in your own words, give one suitable construction use, and state one inspection or safety point.

M3.3 — Substructure and site layout

Substructure and site layout

> Official syllabus focus: Job layout and site clearance for substructure construction.

The substructure transfers building load to the ground. Work begins with understanding the drawing, clearing the site, establishing benchmarks and reference lines, arranging safe access and identifying underground or existing services.

Study method

Protect reference points and check level control before excavation begins.

Construction application

Accurate site preparation avoids misplaced foundations and costly rework.

Technician check

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Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain substructure and site layout in your own words, give one suitable construction use, and state one inspection or safety point.

M3.4 — Setting out load-bearing structures

Setting out load-bearing structures

Official syllabus focus: Center line and face line method for load-bearing structures.

Center-line and face-line methods convert drawings into physical control lines on the ground. Centre lines establish the axes of walls; face lines help establish the actual wall edges. Offsets and diagonal checks are used to preserve right angles and dimensions.

Study method

Measure from a fixed baseline and verify diagonals before digging.

Construction application

Correct setting out ensures the wall thickness and room geometry match the drawing.

Technician check

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Common mistake to avoid

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One-minute recall: Explain setting out load-bearing structures in your own words, give one suitable construction use, and state one inspection or safety point.

M3.5 — Setting out framed structures

Setting out framed structures

> Official syllabus focus: Center line and face line method for framed structures.

For a framed structure, grid or centre lines locate columns and foundations. Face lines then help place formwork and associated elements. Precision is important because column errors propagate into beams, slabs, partitions and services.

Study method

Use independent checks from more than one reference direction rather than trusting a single measurement.

Construction application

Frame layout accuracy is essential for safe load transfer and coordinated construction.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain setting out framed structures in your own words, give one suitable construction use, and state one inspection or safety point.

M3.6 — Earthwork

Earthwork

> Official syllabus focus: Excavation for foundation, timbering and strutting.

Excavation creates the required formation level for foundations. Soil can collapse, especially in deeper or wet excavations, so timbering and strutting provide temporary support where needed. Water control, access and edge protection are essential site-safety concerns.

Study method

Never enter an unsupported trench merely because it appears stable; inspect ground and support conditions continuously.

Construction application

Safe earthwork protects workers and maintains the correct foundation formation.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□□ □□□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□.

One-minute recall: Explain earthwork in your own words, give one suitable construction use, and state one inspection or safety point.

M3.7 — Plinth filling

Plinth filling

> Official syllabus focus: Materials used for plinth filling.

Plinth filling brings the internal ground level to the required formation level using suitable material placed and compacted in layers. Material should be selected for stability, drainage behaviour and compaction response rather than simply for low cost.

Study method

Place and compact in controlled layers; uncontrolled dumping can lead to settlement and floor cracking.

Construction application

Good plinth filling supports durable ground floors and prevents differential settlement.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□.

One-minute recall: Explain plinth filling in your own words, give one suitable construction use, and state one inspection or safety point.

M3.8 — Foundations

Foundations

> Official syllabus focus: Types of foundations: shallow and deep foundations.

Foundations distribute structural load safely into the ground. Shallow foundations are used where suitable bearing stratum is near the surface; deep foundations transfer load to deeper strata or use skin friction and end bearing. Ground condition, load, settlement and nearby construction guide the choice.

Study method

Do not select a foundation type by building height alone; soil and load path must be considered together.

Construction application

Foundation selection is a direct connection between geotechnical condition and building safety.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□ □□□□□□□ □□□□ □□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□.

One-minute recall: Explain foundations in your own words, give one suitable construction use, and state one inspection or safety point.

M3.9 — Pile foundations

Pile foundations

> Official syllabus focus: Types of pile foundation.

Piles are slender deep foundation elements used where near-surface soil is weak, loads are high or settlement control is important. They may transfer load through end bearing, skin friction or both. Installation method and quality control matter as much as the nominal pile type.

Study method

Keep pile records, position checks and testing information because hidden work must be documented.

Construction application

Piles are common below heavy structures and difficult ground conditions.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□ material □□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□ □□□□□□□ □□□□ □□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□.

One-minute recall: Explain pile foundations in your own words, give one suitable construction use, and state one inspection or safety point.

M3.10 — Well foundations

Well foundations

Official syllabus focus: Types of well foundation.

Well foundations are deep foundation systems commonly associated with bridge and water-related works. They are sunk to a required level and require control of position, tilt and scour-related design considerations.

Study method

Treat sinking, alignment and foundation-depth control as linked operations rather than isolated tasks.

Construction application

Well foundations illustrate how foundation form responds to water, depth and lateral effects.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

Malayalam support: □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□.

One-minute recall: Explain well foundations in your own words, give one suitable construction use, and state one inspection or safety point.

M3 closing checklist

- ✓ I can explain prefabrication and basic structural-steel building methods.
- ✓ I can follow the logical sequence from site clearance to setting out and excavation.
- ✓ I can choose a foundation concept from ground and loading conditions.
- ✓ I can draw or describe one site-relevant application and one likely mistake.

M4 • CO4 • 15 HOURS

Building components and roofing

Understand building openings, movement systems and roofs as functional assemblies that must control access, light, air, water and load.

Learning goals

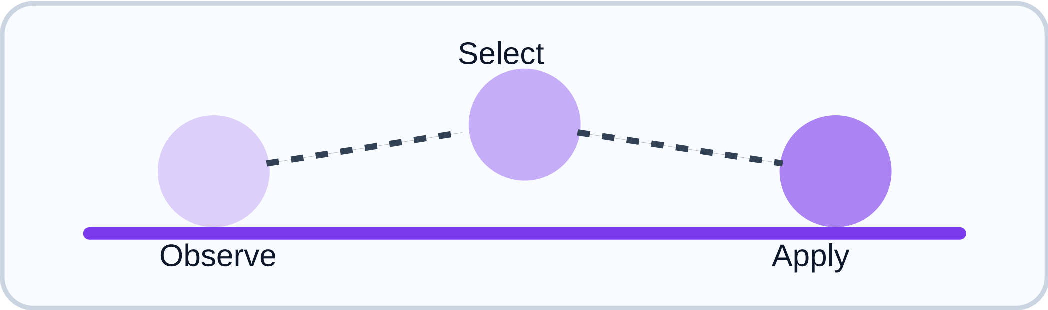
- ✓ Identify common doors, windows, ventilators and their functions.
- ✓ Explain stairs, lifts and escalators at an introductory building-components level.
- ✓ Compare flat and pitched roofs and recognise roof-truss terminology.

Engineering habit

Observe the service condition first. Then identify material or component properties, construction detail, inspection point, safety condition and maintenance consequence.

Exam habit

Start with the official term, define it clearly, classify or list the required points, use a labelled sketch where useful, and conclude with a relevant construction application.



Use the sequence: observe the requirement, select the suitable component or material, then apply it with correct site practice.

M4.1 — Doors

Doors

> Official syllabus focus: Fully panelled, partly panelled, flush and collapsible doors.

Door selection considers privacy, appearance, fire and security needs, durability and location. Panelled doors use framed panels, flush doors use a flat face construction, and collapsible doors provide space-efficient closing in suitable commercial situations.

Study method

Check swing, clearance, frame fixing and exposure before selecting a door type.

Construction application

Correct door selection supports circulation, security and serviceability.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□□ □□□□□□□□□□□□□□□□, □□□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□□ □□□□□□□ □□□□□□□□□□ □□□□□□ □□□□□□□□□□.

One-minute recall: Explain doors in your own words, give one suitable construction use, and state one inspection or safety point.

M4.2 — Doors and shutters

Doors and shutters

> Official syllabus focus: Rolling shutters, revolving doors and glazed doors.

Rolling shutters protect shop and service openings; revolving doors manage entry flow in suitable public buildings; glazed doors provide visibility and light while demanding safe glass selection and hardware. Each type has operational and safety implications.

Study method

Choose hardware, glass and emergency egress arrangements as a system, not as decoration.

Construction application

Door operation affects both daily use and safe movement during emergencies.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□.

One-minute recall: Explain doors and shutters in your own words, give one suitable construction use, and state one inspection or safety point.

M4.3 — Windows

Windows

> Official syllabus focus: Fully panelled, partly panelled and glazed windows; wooden, steel and aluminium windows.

Windows provide light, view, ventilation and weather protection. Frame material influences maintenance, strength, corrosion resistance and appearance. Glazing and opening arrangement must suit the room, climate and cleaning access.

Study method

Consider water drainage, sill detail and movement of the frame when choosing a window system.

Construction application

Good window selection improves daylight, ventilation and building comfort.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□.

One-minute recall: Explain windows in your own words, give one suitable construction use, and state one inspection or safety point.

M4.4 — Window forms

Window forms

> Official syllabus focus: Sliding, louvered, bay, corner, gable and dormer windows; skylight and ventilators.

Different forms address different design needs. Sliding windows save swing space, louvered windows manage ventilation and privacy, bay and corner windows change view and daylight, while gable, dormer and skylight openings relate to roof geometry. Ventilators remove warm or stale air from high positions.

Study method

For every opening, decide where water will go and how the opening can be cleaned and maintained.

Construction application

Window form connects architectural expression with ventilation, light and weather control.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain window forms in your own words, give one suitable construction use, and state one inspection or safety point.

Stairs

Official syllabus focus: Types of stairs and terms used in stairs.

Stairs provide safe vertical movement through an ordered sequence of risers, treads, landings, flights, handrails and supporting members. Terms must be used precisely because they communicate dimensions, safe movement and drawing requirements.

Study method

Keep rise and tread consistent within a flight and provide safe landings and handrails.

Construction application

Stair knowledge is essential in building drawings, access planning and construction supervision.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

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One-minute recall: Explain stairs in your own words, give one suitable construction use, and state one inspection or safety point.

M4.6 — Lifts and escalators

Lifts and escalators

> Official syllabus focus: Lifts and escalators.

Lifts and escalators are mechanised vertical-transport systems. At this level, understand their purpose, basic arrangement and the need for shafts, pits, machine/service spaces, power, safety devices and regular maintenance.

Study method

Coordinate building openings and service requirements early; later changes are difficult and expensive.

Construction application

Vertical transport improves access and passenger movement in multi-storey and public buildings.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

Malayalam support: □ □□□□□□ □□□□□□□□□□□□□□ material □□□□□□□□□□□□ component □□□□□□ □□□□□□□□□□□□□□□□, □□□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□.

One-minute recall: Explain lifts and escalators in your own words, give one suitable construction use, and state one inspection or safety point.

Roofs and roof trusses

> Official syllabus focus: Flat roof, pitched roof and terms used in roofs.

A roof completes the weather envelope. Flat roofs require positive drainage even when visually level; pitched roofs shed water through slope and covering. Roof terms describe members, slopes, edges and drainage elements so that drawings and workmanship can be communicated clearly.

Study method

Trace rainwater from roof surface to outlet; any interrupted route can create leakage or deterioration.

Construction application

Roof understanding helps prevent water ingress and supports correct truss and covering decisions.

Technician check

Before approving or explaining this item, identify the service condition, expected load or exposure, workmanship requirement, inspection point and likely consequence of a poor choice.

Common mistake to avoid

Do not select a material or construction method from appearance alone. Relate the decision to moisture, load, durability, detail, safety and maintenance.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□ material □□□□□□□□□□□ component □□□□□ □□□□□□□□□□□□□□□, □□□□□ service condition □□□, □□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□.

One-minute recall: Explain roofs and roof trusses in your own words, give one suitable construction use, and state one inspection or safety point.

M4 closing checklist

- ✓ I can identify common doors, windows, ventilators and their functions.
- ✓ I can explain stairs, lifts and escalators at an introductory building-components level.
- ✓ I can compare flat and pitched roofs and recognise roof-truss terminology.
- ✓ I can draw or describe one site-relevant application and one likely mistake.

Materials and construction bank

This bank condenses selection logic. It is not a replacement for the detailed module explanations.

Quick comparison of important materials and systems

Item	Key idea	Suitable use / advantage	Watch point
Stone	Natural rock; quality depends on soundness and exposure.	Masonry, foundation, finish.	Cracks, bedding, absorption and weathering.
Timber	Grain-based natural material requiring seasoning.	Joinery, doors, temporary works.	Moisture, defects, pests and fire protection.
Burnt clay brick	Manufactured masonry unit.	General wall construction.	Uniform size, burning quality and joints.
AAC block	Lightweight aerated block.	Lightweight wall systems.	Compatible mortar and careful handling.
Fly ash brick	Controlled industrial-by-product unit.	Masonry where specification permits.	Product quality and curing.
Glass	Transparent or specialised glazing material.	Windows, partitions, daylight.	Safety glass, edges and framing.
Waterproofing	Layer/system resisting water entry.	Roofs, wet areas and foundations.	Continuity, laps and damage protection.
Piles	Deep slender foundation elements.	Weak near-surface soil or heavy loads.	Installation, records and quality checks.
Roof	Weather-protection assembly.	Top building envelope.	Slope, drainage and flashing/detailing.

Visual revision link: M1

Return to the module diagram and explain the observe → select → apply sequence for the listed material or building component.

Open M1 visual lesson

Visual revision link: M2

Return to the module diagram and explain the observe → select → apply sequence for the listed material or building component.

Open M2 visual lesson

Visual revision link: M3

Return to the module diagram and explain the observe → select → apply sequence for the listed material or building component.

Open M3 visual lesson

Visual revision link: M4

Return to the module diagram and explain the observe → select → apply sequence for the listed material or building component.

Open M4 visual lesson

Suggested student activities for self-learning

The following activities are reproduced from the supplied source. Each has 6 hours; total self-learning time is 90 hours. Document observations, photographs only where permission is obtained, measurements, references and a short conclusion.

Official self-learning activity plan

Week	Activity	Hours	Evidence to submit
1	Study rock classification and prepare a report on sustainable materials used in construction.	6	Observation/report/sketch/comparison with clear source acknowledgement.
2	Case study: bamboo and timber in low-cost or rural construction.	6	Observation/report/sketch/comparison with clear source acknowledgement.
3	Compare burnt clay bricks, fly ash bricks and AAC blocks.	6	Observation/report/sketch/comparison with clear source acknowledgement.
4	Collect natural building-material samples and prepare a report.	6	Observation/report/sketch/comparison with clear source acknowledgement.

Week	Activity	Hours	Evidence to submit
5	Suggest flooring for residential, hospital, educational and industrial buildings, with reasons.	6	Observation/report/sketch/comparison with clear source acknowledgement.
6	Visit a hardware shop to identify tiles, glass, ACP, PVC boards, plywood, MDF, HDF and particle board.	6	Observation/report/sketch/comparison with clear source acknowledgement.
7	Measure flooring material in a home and estimate its cost.	6	Observation/report/sketch/comparison with clear source acknowledgement.
8	Watch a video on building glass types and record the use of each type.	6	Observation/report/sketch/comparison with clear source acknowledgement.
9	Prepare a neat sketch of centre-line and face-line methods.	6	Observation/report/sketch/comparison with clear source acknowledgement.
10	Prepare an excavation estimate for a neighbourhood foundation.	6	Observation/report/sketch/comparison with clear source acknowledgement.
11	Watch prefabricated-construction videos and summarise advantages.	6	Observation/report/sketch/comparison with clear source acknowledgement.
12	Select foundation types for a bridge pier, multi-storey building and black-cotton soil.	6	Observation/report/sketch/comparison with clear source acknowledgement.
13	Study arches: definition, types and materials used.	6	Observation/report/sketch/comparison with clear source acknowledgement.
14	Identify fixtures and fastenings for doors and windows.	6	Observation/report/sketch/comparison with clear source acknowledgement.
15	Record BIS-recommended door and window sizes from an approved source.	6	Observation/report/sketch/comparison with clear source acknowledgement.

Field and shop-visit safety

Do not climb on materials, enter excavation, operate equipment or collect samples without permission and supervision. Observe from a safe distance and use photographs only where allowed.

Assessment methodology and module-wise questions

The assessment figures and pattern below follow the source PDF. The questions and answers are study support, not official questions.

Official theory assessment methodology

Component	Mode	Marks / converted marks	Tentative timing
CA1	Attendance	15 / 5	End of semester

Component	Mode	Marks / converted marks	Tentative timing
CA2–CA5	Self-learning activities for Modules 1–4	15 each; converted within CIE	By weeks 4, 7, 11 and 14
CA6	Series examination (2 modules)	50 / 20	Week 8
Series examination	Series examination (2 modules)	50 / 20	Week 15
ESE	Written theory examination, 2.5 hours	60	End of semester

Official examination pattern summary

Examination	Part A	Part B	Part C	Total
Series exam (2 hours)	4 × 1 mark	6 × 3 marks	4 × 7 marks with choice	50
End-semester exam (2.5 hours)	2 questions per module, 1 mark each = 8	2 out of 3 per module, 3 marks each = 24	1 set with choice per module, 7 marks each = 28	60

Module 1 questions with answers

Q1.1. Classify building materials with one example for each class.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.2. State the requirements of a good building stone.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.3. Explain quarrying and dressing of stone.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.4. Describe timber seasoning and state its purpose.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.5. List common timber defects and their effect on use.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.6. Explain the constituents and manufacture stages of burnt clay brick.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.7. Differentiate fly ash brick and AAC block at an introductory level.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q1.8. Describe a field test on bricks and state what it indicates.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Module 2 questions with answers

Q2.1. Compare granite, ceramic tile and pavement block for suitable uses.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.2. Differentiate plywood, particle board and veneer.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.3. State uses of toughened and borosilicate glass.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.4. Differentiate ferrous and non-ferrous metals with construction examples.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.5. Explain how waterproofing differs from thermal insulation.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.6. State uses and handling considerations of POP.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.7. Describe the construction relevance of fly ash, slag, rice husk and coir fibre.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q2.8. Explain MDF, HDF, WPC, ACP and M-sand in brief.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Module 3 questions with answers

Q3.1. Define prefabricated structure and state advantages.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.2. Outline the construction sequence of a structural-steel building.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.3. Explain site clearance and its purpose.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.4. Describe centre-line and face-line setting out.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.5. State safety points during foundation excavation.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.6. Explain the need for timbering and strutting.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.7. Differentiate shallow and deep foundations.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q3.8. State basic uses of pile and well foundations.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Module 4 questions with answers

Q4.1. Differentiate panelled, flush and collapsible doors.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.2. State uses of rolling shutters and glazed doors.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.3. Compare wooden, steel and aluminium windows.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.4. Explain the function of louvered, bay and dormer windows.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.5. Define riser, tread, flight and landing.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.6. State the purpose of lifts and escalators in buildings.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.7. Differentiate flat and pitched roofs.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Q4.8. List roof terms that must be understood while reading a roof drawing.

Answer approach: begin with the correct technical term, organise the response into definition, classification or procedure as required, then add one practical construction implication. A high-mark answer should include a labelled sketch or table whenever it improves clarity.

For this course, avoid unsupported numerical values. Use the exact language of the question and connect the answer to material selection, site practice, safety, durability or maintenance.

Practice model question paper

This is a practice paper based on the official 60-mark pattern in the supplied syllabus. It is not an official SITTTR model question paper.

POLY PMNA PRACTICE PAPER — Course 4032 Electrical Installation Design and Estimation
Revision 2021 library placement • Duration: 2.5 hours • Maximum marks: 60

PART A — Answer all questions (8 × 1 = 8)
1. State one requirement of a good building stone.
2. Name one method of seasoning timber.
3. State one use of fly ash brick or AAC block.
4. Name one type of insulating material/protection.
5. Define prefabricated structure.
6. Distinguish shallow and deep foundation in one point.
7. Define tread.
8. State one purpose of a ventilator.

PART B — Answer any two questions from each module (4 × 6 = 24)
Module I: Explain brick manufacturing; explain timber defects; compare stone and laterite selection.
Module II: Compare flooring materials; explain wood products; explain insulating/protective materials.
Module III: Explain centre-line setting out; explain excavation safety; compare pile and well foundations.
Module IV: Compare doors; explain window forms; explain flat and pitched roofs.

PART C — Answer one set from each module (4 × 7 = 28)
M1: (a) Prepare a material-selection note for a wall using stone, brick and AAC. OR (b) Explain stone, timber, lime and brick-quality checks.
M2: (a) Select finishes for a hospital room and justify. OR (b) Explain glass, metal, POP and special processed materials.
M3: (a) Describe the sequence from site clearance to foundation. OR (b) Explain prefabrication, steel construction and deep foundations.
M4: (a) Explain doors, windows and ventilators as functional building components. OR (b) Explain stairs, lifts, escalators and roof systems.

Practice-paper answer guide

Use this guide to check coverage and structure. In the examination, follow the exact question and marks allocation.

Part A

Give a one-sentence correct definition, property, use or distinction. Do not add unrelated discussion.

Part B

Use a short introduction, 4–6 organised technical points, and a simple sketch/table where it improves the answer. Include use and a practical caution.

Part C

Plan the answer before writing. Begin with definition/scope, organise subheadings in sequence, show selection criteria or construction sequence, include labelled diagrams, then conclude with safety/durability implications.

M1 high-mark framework

A strong answer should use the module terms: Classification of materials, Stone, Stone quality, Quarrying and dressing of stone, Timber, Seasoning of timber, Defects and bamboo, Lime, Brick earth and traditional bricks, Burnt clay bricks, Fly ash bricks and AAC blocks. State the service requirement, explain the construction/material logic, add a labelled sketch or comparison, then finish with an application and one caution.

M2 high-mark framework

A strong answer should use the module terms: Flooring materials, Wood products, Glass, Metals, Insulating and protective materials, Plaster of Paris, Industrial and agro-waste materials, Special processed materials. State the service requirement, explain the construction/material logic, add a labelled sketch or comparison, then finish with an application and one caution.

M3 high-mark framework

A strong answer should use the module terms: Prefabricated structures, Structural steel buildings, Substructure and site layout, Setting out load-bearing structures, Setting out framed structures, Earthwork, Plinth filling, Foundations, Pile foundations, Well foundations. State the service requirement, explain the construction/material logic, add a labelled sketch or comparison, then finish with an application and one caution.

M4 high-mark framework

A strong answer should use the module terms: Doors, Doors and shutters, Windows, Window forms, Stairs, Lifts and escalators, Roofs and roof trusses. State the service requirement, explain the construction/material logic, add a labelled sketch or comparison, then finish with an application and one caution.

Quick revision, glossary and references

Use the cards below in the final revision session. Then return to module details for any uncertain topic.

M1 quick revision

Classify materials and make sound first-stage choices for stone, timber, lime, bricks, blocks and alternative masonry units.

✓ Classification of materials: Types: natural, artificial, special, finished and recycled.

- ✓ Stone: Classification of rocks: igneous, sedimentary and metamorphic.
- ✓ Stone quality: Requirements of good building stone; general characteristics; laterite stones.
- ✓ Quarrying and dressing of stone: Quarrying and dressing of stone.

M2 quick revision

Select surface, board, glass, metal, insulating and special materials by service condition rather than appearance alone.

- ✓ Flooring materials: Natural and artificial flooring: granite, marble, clay tiles, ceramic tiles and pavement blocks.
- ✓ Wood products: Plywood, particle board, veneers, laminated board and their uses.
- ✓ Glass: Soda lime, lead, toughened and borosilicate glass: types and uses.
- ✓ Metals: Ferrous and non-ferrous metals and their uses.

M3 quick revision

Move from material knowledge to the organised construction of a safe substructure, including layout, excavation and foundation choice.

- ✓ Prefabricated structures: Brief description of prefabricated structures.
- ✓ Structural steel buildings: Construction methods for structural steel buildings.
- ✓ Substructure and site layout: Job layout and site clearance for substructure construction.
- ✓ Setting out load-bearing structures: Center line and face line method for load-bearing structures.

M4 quick revision

Understand building openings, movement systems and roofs as functional assemblies that must control access, light, air, water and load.

- ✓ Doors: Fully panelled, partly panelled, flush and collapsible doors.
- ✓ Doors and shutters: Rolling shutters, revolving doors and glazed doors.
- ✓ Windows: Fully panelled, partly panelled and glazed windows; wooden, steel and aluminium windows.
- ✓ Window forms: Sliding, louvered, bay, corner, gable and dormer windows; skylight and ventilators.

Glossary

Term	Meaning for this course
AAC	Autoclaved aerated concrete; a lightweight aerated masonry unit.
AAC block joint	The specified jointing/mortar arrangement needed for a block system.
Bearing stratum	Soil layer capable of receiving foundation load.
Centre line	Reference line used to position walls, columns or foundations.

Term	Meaning for this course
Dressing	Shaping stone to the required size and surface.
Face line	Line representing the actual face/edge of a wall or member in setting out.
Fly ash	Fine industrial by-product used in appropriate construction products.
Plinth filling	Compacted filling used to bring an internal level to the required formation.
POP	Plaster of Paris; gypsum-based interior finishing material.
Prefabrication	Producing building components before site erection.
Seasoning	Controlled moisture reduction in timber.
Shallow foundation	Foundation that transfers load near ground surface.
Timbering and strutting	Temporary support to the sides of an excavation.
Ventilator	High-level opening supporting air movement.
WPC	Wood-polymer composite board/product used for specified applications.

Officially listed learning resources

Books and web resources named in the supplied source

Type	Resource
Source list	Building Construction — S. C. Rangawala — Charotar Publication.
Source list	Construction Materials — D. N. Ghose — Tata McGraw Hill.
Source list	Engineering Materials — S. C. Rangwala — Charotar Publication.
Source list	Laboratory Manual on Testing of Engineering Materials — H. Sood — New Age Publishers.
Source list	KBR; BIS (as listed in the PDF).
Source list	
Source list	
Source list	